

12 A

In moving from Q to P,

$$\frac{\phi_Q}{\phi_P} = \frac{r_P}{r_Q}$$

$$\begin{aligned}\phi_Q &= \frac{r}{2r}(-800) \\ &= -400 \text{ kJ kg}^{-1}\end{aligned}$$

$$\begin{aligned}\Delta GPE &= m\Delta\phi \\ &= 1.0[-800 - (-400)] \\ &= -400 \text{ kJ}\end{aligned}$$

13 D